

Q1 3. Identify an errors in the following snippet

```
int binarySearch(int arr[], int n, int key)
{
    int low = 0, high = n - 1;
    while(low < high)
    { int mid = (low + high) / 2;
      if(arr[mid] == key) return mid;
      else if(arr[mid] < key)
      low = mid;
      else high = mid;
    }
    return -1;
}
```

Your Code

```
1 int binarySearch(int arr[], int n, int key)
2 {
3     int low = 0, high = n - 1;
4
5     while(low <= high)
6     {
7         int mid = (low + high) / 2;
8
9         if(arr[mid] == key) return mid;
10        else if(arr[mid] < key)
11            low = mid + 1;
12        else
13            high = mid - 1;
14    }
15
16    return -1;
17 }
```

Input

stdin input

Output

```
Compilation Error:
C:/CCPPComp/CCPP1/bin/./lib/gcc
/x86_64-w64-
mingw32/10.3.0/././././././x86_6
4-w64-mingw32/bin/ld.exe:
C:/CCPPComp/CCPP1/bin/./lib/gcc
```

Q2 8. Buggy code in node creation

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

int main() {
    struct Node *newNode;

    newNode->data = 10;
    newNode->next = NULL;

    printf("%d", newNode->data);

    return 0;
}
```

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 struct Node {
4     int data;
5     struct Node *next;
6 };
7 int main() {
8     struct Node *newNode;
9
10    newNode = (struct Node*)malloc(sizeof(struct Node));
11    if (newNode == NULL) {
12        printf("Memory allocation failed");
13        return 1;
14    }
15    newNode->data = 10;
16    newNode->next = NULL;
17
18    printf("%d", newNode->data);
19    free(newNode);
20    return 0;
21 }
```

stdin input

Output

10

Q3 1. Find the bugs, include the missing lines in the following snippet.

```
int linearSearch(int arr[], int n, int key) {  
    for(int i = 0; i <= n; i++) {  
        if(arr[i] == key)  
            return i;  
    }  
    return -1;  
}
```

```
1 int linearSearch(int arr[], int n, int key) {  
2     for(int i = 0; i < n; i++) {  
3         if(arr[i] == key)  
4             return i;  
5     }  
6     return -1;  
7 }
```

stdin input

Output

```
Compilation Error:  
C:/CCPPComp/CCPP2/bin/./lib/gcc/  
/x86_64-w64-  
mingw32/10.3.0/../../../../x86_64-  
w64-mingw32/bin/ld.exe:  
C:/CCPPComp/CCPP2/bin/./lib/gcc/
```